Experiment No : 6

Title: Implement Stack and Queue using Linked List.

Name of Student: Shamiksha Chaudhari
Class: SY **Div:** SY1 **PRN No:** B25ET1005 **Batch:** A1

Program:

PART 1
Stack using linked list

```
#include <stdio.h>
#include <stdlib.h>
struct Node
{
   int data ;
   struct Node* next ;
};

struct Node*top = NULL;
void push(int value)
{
   struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
   if (newNode == NULL)
   {
      printf("Stack is full and memory allocation failed\n");
      return;
   }
   newNode->data = value ;
   newNode->next = top;
   top = newNode;
   printf("%d pushed to stack\n", value);
}
```

```
void pop()
{
   if (top == NULL)
   {
      printf("Stack is empty \n");
      return;
   }
   struct Node* temp = top;
   printf("%d popped from stack\n", top->data);
   top = top->next;
   free(temp);
}


void peek() {
   if (top == NULL)
   {
      printf("Stack is empty\n");
      return;
   }
   printf("Top element is %d\n", top->data);
}


void display()
{
   if (top == NULL)
   {
      printf("Stack is empty\n");
      return;
   }
   struct Node* temp = top;
```

```
    printf("Stack: ");
    while (temp != NULL)
    {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}


int main()
{
    int choice, value;

    while (1)
    {
        printf("");
        printf("\n1. Push\n2. Pop\n3. Peek\n4. Display\n5. Exit\n");
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
            case 1:
                printf("Enter value to push: ");
                scanf("%d", &value);
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:
                peek();
```

```
            break;
        case 4:
            display();
            break;
        case 5:
            exit(0);
        default:
            printf("Invalid choice!\n");
    }
  }

  return 0;
}
```

OUTPUT:

```
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value to push: 10
10 pushed to stack

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value to push: 20
20 pushed to stack

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value to push: 30
30 pushed to stack
```

```
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
30 popped from stack
```

```
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Top element is 20
```

```
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: 20 10
```

```
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
```

PART2

Queue using linked list

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
int data;
struct Node* next;
};

struct Node* front = NULL;
struct Node* rear = NULL;

void enqueue(int value) {
struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
newNode->data = value;
newNode->next = NULL;
if (rear == NULL) {
front = rear = newNode;
} else {
rear->next = newNode;
rear = newNode;
```

```
}
printf("%d enqueued to queue\n", value);
}

void dequeue() {
if (front == NULL) {
printf("Queue is empty\n");
return;
}
struct Node* temp = front;
printf("%d dequeued from queue\n", front->data);
front = front->next;
if (front == NULL) // if queue becomes empty
rear = NULL;
free(temp);
}

void display() {
if (front == NULL) {
printf("Queue is empty\n");
return;
}
struct Node* temp = front;
printf("Queue: ");
while (temp != NULL) {
printf("%d ", temp->data);
temp = temp->next;
}
printf("\n");
}

int main() {
```

```
enqueue(10);
enqueue(20);
enqueue(30);
display();
dequeue();
display();
enqueue(40);
display();
return 0;
}
```